

# Boyan LI

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## Education

Hong Kong University of Science and Technology (Guangzhou)

Guangzhou, China

Ph.D. Student in Data Science and Analytics (DSA)

Sep. 2024 – Present

Supervisor: Prof. YuYu LUO

Southern University of Science and Technology (SUSTech)

Shenzhen, China

B.Eng. in Computer Science and Technology

Sep. 2019 – Jul. 2023

GPA: 3.91/4.0 (Rank 2/183) | Honors: Outstanding Graduate, Highest Honors

## Publications

\* indicates equal contribution. **Boyan Li** is bolded.

### First-Author Publications (Including Co-First)

- [1] **DataSpace: Benchmarking Data Agents for Verifiable Analytics over Heterogeneous Workspaces**  
**Boyan Li\***, Z. Liang\*, Y. Xie\*, X. Lin, T. Luo, X. Liu, Y. Zhu, Z. Peng, Y. Li, Z. Zhang, J. Zhang, N. Tang, G. Li, Y. Luo  
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2027 (Under Review)  
Official evaluation benchmark for the KDD Cup 2026 Data Agents for Complex Data Analysis competition.
- [2] **DeepEye-SQL: A Software-Engineering-Inspired Text-to-SQL Framework**  
**Boyan Li**, C. Chen, Z. Xue, Y. Mei, Y. Luo  
*ACM SIGMOD International Conference on Management of Data (SIGMOD)*, 2026 (CCF-A)
- [3] **DeepEye: A Steerable Self-driving Data Agent System**  
**Boyan Li**, Y. Peng, Y. Xie, S. Lu, Y. Zhu, X. Mu, X. Liu, Y. Luo  
*ACM SIGMOD International Conference on Management of Data (SIGMOD)*, 2026 (Demo, CCF-A)
- [4] **DeepEye: A Workflow-Centric Agentic Data System for Steerable Data Analytics**  
**Boyan Li**, Y. Peng, Y. Xie, S. Lu, Y. Zhu, X. Mu, X. Liu, J. Tang  
*VLDB Workshop on Agentic Data Systems and Data-Centric AI (ADS/DATAI)*, 2026  
**Best Paper Award.**
- [5] **DPC: Training-Free Text-to-SQL Candidate Selection via Dual-Paradigm Consistency**  
**Boyan Li**, O. O. K. Hei, Y. Yu, Y. Luo  
*Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026 (Main Conference, CCF-A)
- [6] **ROSE: An Intent-Centered Evaluation Metric for NL2SQL**  
W. Pei\*, S. Hou\*, **Boyan Li\***, H. Chen, Z. Shi, Y. Luo  
*Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026 (Main Conference, CCF-A)
- [7] **Alpha-SQL: Zero-Shot Text-to-SQL using Monte Carlo Tree Search**  
**Boyan Li**, J. Zhang, J. Fan, Y. Xu, C. Chen, N. Tang, Y. Luo  
*International Conference on Machine Learning (ICML)*, 2025 (CCF-A)  
Deployed in **Huawei Cloud DataArts Insight** and **Wuhan Urban Operation Agent**.
- [8] **DeepVIS: Bridging Natural Language and Data Visualization Through Step-wise Reasoning**  
Z. Shuai\*, **Boyan Li\***, S. Yan, Y. Luo, W. Yang  
*IEEE VIS Conference (VIS)*, 2025 (CCF-A)
- [9] **The Dawn of Natural Language to SQL: Are We Fully Ready?**  
**Boyan Li**, Y. Luo, C. Chai, G. Li, N. Tang  
*Proceedings of the VLDB Endowment (VLDB)*, 2024 (CCF-A)  
VLDB 2024 Top-2 Highly Cited Paper (Google Scholar Metrics).

### Co-authored Publications

- [10] **MarkovSQL: Adaptive Text-to-SQL Workflow Orchestration Under Partial Observability**  
Y. Zhu, **Boyan Li**, Y. Luo  
*ACM SIGMOD International Conference on Management of Data (SIGMOD)*, 2027 (Under Review)
- [11] **SQLConductor: Search-to-Policy Learning for Step-wise Text-to-SQL Orchestration**  
Y. Zhu, Z. Peng, **Boyan Li**, Y. Luo  
*Proceedings of the VLDB Endowment (VLDB)*, 2027 (Under Review)
- [12] **CoEvo-SQLi: Co-Evolving SQL Injection Detectors via Schema-Grounded Attack Synthesis**  
H. Pan, X. Lin, **Boyan Li**, Y. Luo  
*International Conference on Learning Representations (ICLR)*, 2027 (Under Review)
- [13] **MIRA: Evidence-Verified Repair Memory for Text-to-SQL Correction**  
Y. Liu, C. Yang, **Boyan Li**, R. Mao, Y. Luo  
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2027 (Under Review)
- [14] **SCOPE-SQL: Structured Semantic Credit for Reinforcement Learning of Interactive SQL Repair Agent**  
H. Qi, J. Zhao, **Boyan Li**, J. Jiang, Y. Luo  
*Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026 (Findings)

- [15] **DataMagic: Transforming Tabular Data into Data Insight Video**  
Y. Xie, C. Ma, Z. Wang, L. Wang, J. Zhu, C. Zeng, Z. Shen, **Boyan Li**, Y. Luo  
*Proceedings of the VLDB Endowment (VLDB)*, 2026 (Demo Track, CCF-A)
- [16] **ExpeSQL: An Efficient, Experience-Guided Decompositional Search Framework for Text-to-SQL**  
Z. Yang, S. Yan, **Boyan Li**, Y. Xu, L. Song, Y. Matsuda, W. Han, B. Bai  
*VLDB Workshop on Agentic Data Systems and Data-Centric AI (ADS/DATAI)*, 2026
- [17] **How to Make Tool-Using LLM Agents Efficient? A Systematic Survey**  
Y. Hu, Z. Cheng, W. Pei, M. Li, **Boyan Li**, Y. Zhu, Y. Xie, H. Chen, S. Hou, Y. Luo, H. Cheng, J. X. Yu, H. Du  
*Conference on Language Modeling (COLM)*, 2026 (Under Review)
- [18] **A Survey of NL2SQL with Large Language Models: Where are we, and where are we going?**  
X. Liu, S. Shen, **Boyan Li**, P. Ma, R. Jiang, Y. Zhang, J. Fan, G. Li, N. Tang, Y. Luo  
*IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 2025 (CCF-A)
- [19] **nvBench 2.0: A Benchmark for Natural Language to Visualization under Ambiguity**  
T. Luo, C. Huang, L. Shen, **Boyan Li**, S. Shen, W. Zeng, N. Tang, Y. Luo  
*Conference on Neural Information Processing Systems (NeurIPS)*, 2025 (CCF-A)
- [20] **NL2SQL-BUGs: A Benchmark for Detecting Semantic Errors in NL2SQL Translation**  
X. Liu, S. Shen, **Boyan Li**, N. Tang, Y. Luo  
*ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2025 (CCF-A)
- [21] **Augmenting Realistic Charts with Virtual Overlays**  
Y. Shi, **Boyan Li**, Y. Luo, L. Chen, N. Tang  
*CHI Conference on Human Factors in Computing Systems (CHI)*, 2025 (CCF-A)
- [22] **EllieSQL: Cost-Efficient Text-to-SQL with Complexity-Aware Routing**  
Y. Zhu, R. Jiang, **Boyan Li**, N. Tang, Y. Luo  
*Conference on Language Modeling (COLM)*, 2025
- [23] **A Survey of Data Agents: Emerging Paradigm or Overstated Hype?**  
Y. Zhu, ..., **Boyan Li**, et al.  
*IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 2025 (Under Review)  
[Ranked Top-3 in Hugging Face Daily Papers.](#)
- [24] **Advances and Challenges in Foundation Agents**  
B. Liu, ..., **Boyan Li**, et al.  
*arXiv preprint*, 2025  
[Ranked Top-1 in Hugging Face Daily Papers.](#)
- [25] **Are Large Language Models Good Statisticians?**  
Y. Zhu, S. Du, **Boyan Li**, Y. Luo, N. Tang  
*Conference on Neural Information Processing Systems (NeurIPS)*, 2024 (CCF-A)
- [26] **A Plug-and-Play Natural Language Rewriter for Natural Language to SQL**  
P. Ma, **Boyan Li**, R. Jiang, J. Fan, N. Tang, Y. Luo  
*arXiv preprint*, 2024

## Open Source & Systems

- **DeepEye: A Steerable Self-driving Data Agent System** [\[GitHub\]](#)
  - Led the development of a low-code orchestration platform for transparent data analysis.
  - **Awards:** “Best Open-source Project Award” at AI Agent 2025 Competition; Silver Medal at the 51st Geneva International Exhibition of Inventions.
- **Alpha-SQL (ICML 2025)** [\[GitHub\]](#)
  - Official implementation of the Monte Carlo Tree Search based Text-to-SQL agent.
  - **Impact:** Deployed in **Huawei Cloud DataArts Insight** and **Wuhan Urban Operation Agent**.
- **NL2SQL Handbook (TKDE 2025)** [\[GitHub\]](#)
  - A comprehensive and continuously updated roadmap for NL2SQL researchers.
  - **Impact:** Widely recognized as a seminal resource for tracking state-of-the-art advancements, with GitHub Stars 1.4k+.
- **NL2SQL360 Evaluation Framework (VLDB 2024)** [\[GitHub\]](#)
  - A comprehensive testbed for fine-grained evaluation of NL2SQL solutions.
  - **Recognition:** Associated with the VLDB 2024 Top-2 Highly Cited Paper.

## Experience

**Tsinghua University (Database Group)**  
Exchange Student (Mentor: Prof. Guoliang Li)  
**Huawei Central Research Institute**  
Research Intern

Beijing, China  
Jun. 2025 – Dec. 2025  
Shenzhen, China  
Jul. 2022 – Sep. 2022

## Honors & Awards

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|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Best Paper Award</b> , VLDB Workshop on Agentic Data Systems (DeepEye)                                                                               | <i>Sep. 2026</i>        |
| <b>Merit Prize</b> , INFH Research Excellence Award                                                                                                     | <i>May. 2026</i>        |
| <b>Silver Medal</b> , 51st Geneva International Exhibition of Inventions (DeepEye Data Agent System)                                                    | <i>Mar. 2026</i>        |
| <b>Silver Medal</b> , 51st Geneva International Exhibition of Inventions (Low-Altitude Economy Ground-Air Collaborative Delivery Digital Twin Platform) | <i>Mar. 2026</i>        |
| <b>Research Excellence Awards</b> , DSA Excellence Awards 2025                                                                                          | <i>Dec. 2025</i>        |
| <b>Runner-up, Best Project Award</b> , DSAA 6000R, HKUST(GZ)                                                                                            | <i>Nov. 2025</i>        |
| <b>Best Open-source Project Award</b> , AI Agent 2025 Competition                                                                                       | <i>Oct. 2025</i>        |
| <b>Merit Prize</b> , DSA Excellent Research Award 2024                                                                                                  | <i>Jan. 2025</i>        |
| <b>Best Project Award</b> , DSAA 5021, HKUST(GZ)                                                                                                        | <i>Dec. 2024</i>        |
| <b>Highest Honors in Computer Science &amp; Engineering</b> , SUSTech                                                                                   | <i>Jun. 2023</i>        |
| <b>Outstanding Graduate</b> , SUSTech                                                                                                                   | <i>Jun. 2023</i>        |
| <b>Huawei Scholarship</b> , Huawei Technologies                                                                                                         | <i>May. 2022</i>        |
| <b>Excellent Student Scholarship</b> , SUSTech                                                                                                          | <i>2020, 2021, 2022</i> |
| <b>National Encouragement Scholarship</b> , Ministry of Education of China                                                                              | <i>Apr. 2020</i>        |

## Services

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| <b>General Chair</b> : Data Agents for Complex Data Analysis (KDD Cup 2026)                                                                                                                                                                                                                                                                                                                               | <i>2026</i>      |
| <b>Program Chair</b> : Agentic Data Systems (VLDB 2026 Workshop)                                                                                                                                                                                                                                                                                                                                          | <i>2026</i>      |
| <b>Conference Reviewer</b> : Neural Information Processing Systems ( <b>NeurIPS</b> ); International Conference on Learning Representations ( <b>ICLR</b> ); ACM SIGKDD Conference on Knowledge Discovery and Data Mining ( <b>KDD</b> ); Annual Meeting of the Association for Computational Linguistics ( <b>ACL</b> ); Conference on Empirical Methods in Natural Language Processing ( <b>EMNLP</b> ) |                  |
| <b>Journal Reviewer</b> : IEEE Transactions on Knowledge and Data Engineering ( <b>TKDE</b> )                                                                                                                                                                                                                                                                                                             |                  |
| <b>Invited Talk</b> : Huawei Squirrel Webinar, “DeepEye-SQL”                                                                                                                                                                                                                                                                                                                                              | <i>Jan. 2026</i> |

## Teaching

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|------------------------------------------------------------------------------------------|-----------------------|
| <b>Teaching Assistant</b> , Future Technology Exploration and Innovation Camp, HKUST(GZ) | <i>Jul. 2026</i>      |
| <b>Teaching Assistant</b> , C++ Programming (UFUG 2601), HKUST(GZ)                       | <i>Spring 2025–26</i> |
| <b>Teaching Assistant</b> , Table Representation Learning (DSAA 6000O), HKUST(GZ)        | <i>Spring 2024–25</i> |